PERSONALIZED LEARNING PLAN · PREPARED JULY 2026

# AI Fluency for Scientific Research

A one-hour-a-day path from AI newcomer to locally-expert practitioner, built for an MD/PhD neuroscientist at the bench.

Foundations → LLMs · Claude & Prompting · Literature Synthesis · No-Code Data Analysis · Scientific Writing · Figures & Slides · Claude Code & Agents

| | |
|---|---|
| Prepared for | **Ezechukwu "Ez" Nduka, MD/PhD Program, Vanderbilt University** |
| Stage | **Entering dissertation research · basic-science neuroscience (wet lab)** |
| Effort budget | **1 focused hour/day · ~6 months** |
| Spend budget | **≤ $200/month (core path uses a fraction of it)** |

Anchored on Anthropic's Claude, with non-Anthropic tools where they fit better.

v1.0

EXECUTIVE SUMMARY

# The plan in one page

This plan takes you from "What is AI?" to being the person in your lab others ask for help applying AI to research. It runs about six months, at one focused hour per day, for well under your $200/month budget.

It is deliberately **hands-on**. Every phase pairs a small amount of high-yield reading, video, or audio with a required exercise done on your own wet-lab material: your papers, your protocols, your spreadsheets, your figures, your writing. Fluency here means doing the task with AI, not watching someone else do it.

### WHAT YOU'LL BE ABLE TO DO

- Explain, in plain language, what an LLM is, how it is trained, and where it fails.
- Run a grounded literature synthesis with zero fabricated citations.
- Analyze a real lab dataset conversationally, producing stats and publication-style plots, without writing code.
- Draft and edit manuscripts and grant text within journal and NIH policy.
- Build figures, schematics, and slide decks quickly, and ethically.
- Use Claude Code and connectors (PubMed, Drive) to automate research chores.

### THE SPEND, DECIDED

**Anchor: Claude Pro at $20/mo.** Verified to cover nearly everything: Projects, Artifacts, Cowork, Claude in Chrome, and the science connectors (PubMed, Benchling, BioRender, Google Drive).

**Add only when a phase needs it:** the BioRender student plan during figure work, and Elicit Plus during a systematic review.

**Optional power-up:** Claude Max ($100/mo), only if you hit usage limits.

A typical month runs **$20 to $55**, so the $200 cap leaves plenty of room.

## Key findings that shaped this plan

- **The "Anthropic science offering" is real, and the useful part is included in Pro.** Anthropic launched *Claude for Life Sciences* in October 2025. Its PubMed, Benchling, and BioRender connectors work on an ordinary $20 Pro plan. The heavier *Claude Science* workbench leans computational and enterprise, so you do not need it as a non-coding wet-lab scientist, and you should not budget extra for it.
- **Claude is a strong synthesis and writing engine, but it is not a citation database.** The most important habit in this plan: never let a chatbot invent references. Discovery happens in grounded tools (Elicit, Semantic Scholar, Scite, NotebookLM), and every citation is verified in Zotero.
- **Image and video generation are non-Anthropic by necessity.** Claude has none. You'll use BioRender for real scientific figures and free tools (Google Gemini, Gamma, Canva) for slides. One hard line applies throughout: never generate anything that could pass as experimental data.
- **No coding required to start.** Because you have no terminal experience yet, Claude Code is deferred to Phase 7 and introduced through a gentle, science-first on-ramp for wrangling PDFs and data, not for building software.

### HOW “EXPERT” IS MADE CONCRETE

There are nine competency domains, each with explicit end-state objectives (Section 2) and a four-level rubric (Novice, Functional, Proficient, Local Expert). Eight milestones (Section 5), one per phase, each ask you to produce and self-score a real artifact. The last is a capstone “research sprint” that chains every skill end to end. That keeps progress visible and self-checkable rather than a matter of feel.

Every resource in this plan was checked against the live web on 8-9 July 2026. Where a specific price or product detail could not be loaded directly (some vendor pages block automated access), it is corroborated from multiple current sources and flagged in Section 10. Consumer-AI prices and model names change monthly, so confirm them at signup.

CONTENTS

# Table of contents

**HOW TO WORK THROUGH THIS**

This document is built to be worked through in order, though it also stands alone as a reference you'll return to. The phases are sequential by design (each assumes the last), while Sections 6-9 are meant to be dipped into anytime. If you ramp faster than expected, the natural place to compress is Phase 1; the high-value core is Phases 3-7. A good rhythm with your mentor is a 15-minute check-in at each milestone (Section 5), using its self-score rubric to keep progress concrete.

SECTION 1

# How to use this plan

One focused hour a day is a real constraint. The plan is built around it: short inputs, immediate practice, and nothing included that isn't high-yield.

## The daily hour

Treat each session as roughly **20 minutes of input** (a video segment, a doc, a short article) and **40 minutes of doing**. The doing is the point. Reading about AI produces the feeling of fluency; using AI on your own material produces the real thing. Where a resource is long, such as a 3-hour Karpathy talk, it is split across sessions.

### RULES THAT MAKE THE HOUR PAY OFF

- **Always use real lab material.** Your papers, your data, your draft. Skip the toy example whenever a real one exists.
- **Keep an AI Lab Notebook.** One running doc: what you tried, the prompt, what worked, what the model got wrong. This becomes your personal playbook.
- **Verify before you trust.** Every fact, number, and citation that leaves a chat and enters your science gets checked against the source.
- **Miss a day, don't miss two.** The schedule has slack built in, and consistency beats intensity.

### DATA HYGIENE FROM DAY ONE

Your PI confirmed that unpublished basic-science data is generally fine in mainstream cloud AI tools. Build good habits anyway:

- Never paste human-subjects or identifiable data into any consumer tool without checking IRB and data-use terms first.
- Strip names, MRNs, and animal-protocol identifiers you don't need.
- Treat anything pre-publication and competitively sensitive with care, and ask your PI when you are unsure.
- Know your setting: consumer Claude does not train on your chats when memory and training controls are off. Confirm your account settings.

## The shape of the six months

Nine phases (0-8) run over about 26 weeks. Each phase has learning objectives, a curated resource shortlist, a hands-on exercise tied to your lab, and a milestone you self-score before moving on. The phases build on each other, and the rubric in Section 2 lets you and your mentor see the climb from novice to local expert in each competency.

| PHASE | FOCUS | WEEKS | ~HOURS | MILESTONE ARTIFACT |
|---|---|---|---|---|
| **0** | Setup & first contact | 1 (part) | 2-3 | Accounts live; first useful chat |
| **1** | Foundations: what AI & LLMs are | 1-3 | 12-14 | 5-minute "explain an LLM" talk |
| **2** | Claude fluency & prompting | 4-5 | 9-10 | Thesis Project + prompt library |

| | | | | |
|---|---|---|---|---|
| **3** | Literature review & synthesis | 6-9 | 18-20 | Verified mini-synthesis, 0 fake cites |
| **4** | Data analysis without coding | 10-12 | 14-15 | Real dataset → stats + figure |
| **5** | Scientific writing & integrity | 13-15 | 14-15 | Edited section + disclosure note |
| **6** | Figures, slides & media | 16-17 | 9-10 | BioRender figure + AI deck |
| **7** | Claude Code, agents & connectors | 18-22 | 22-25 | PDF folder → evidence table |
| **8** | Integration, capstone & habit | 23-26 | 16-18 | End-to-end research sprint |

## Why six months?

At about 5 sessions a week, six months is roughly **120 to 130 focused hours** of real practice on real work. That is the horizon the objectives in Section 2 justify:

- **A 30-day crash course can't reach expertise.** Breadth this wide (foundations, synthesis, analysis, writing, figures, agents) needs spaced practice to stick, and each skill has to be exercised on genuine lab tasks, which take time to arise.
- **Longer than about 8 months works against you.** Tools and model versions drift on a monthly cadence, so momentum and retention fade. Six months is long enough for depth and durable habits, and short enough that what you learn is still current when you finish.
- **It fits your science calendar.** The phases map onto what a second-to-third-year PhD is already doing, such as reading in, generating pilot data, and writing a first-year committee document or F30/F31 aims, so the practice is never contrived.

### IF LIFE COMPRESSES THE TIMELINE

The plan is modular. To still call yourself fluent, the realistic minimum is Phases 0-5 (about 10-12 weeks), which cover foundations, Claude, literature, data, and writing: the daily-driver skills. Phases 6-8 add figures, agentic automation, and integration, which move you from fluent to the person others ask. Whatever you do, keep the milestones. They are the proof.

SECTION 2

# The destination: what "locally expert" means

"Expert" is easy to say and hard to measure. Here it means nine concrete competency domains, each with end-state objectives you should be able to demonstrate, plus a rubric that turns the vague endpoint into a visible climb.

The working definition: **you are locally expert when you can do each task below reliably on your own work, show a labmate how, and know the limits and failure modes well enough to keep the lab out of trouble.** Teaching-readiness and risk-awareness are what separate a good user from a local expert.

## The nine competency domains & their objectives

**D1 · Foundational literacy.** Explain in plain language, to a scientist or a family member, what AI, machine learning, deep learning, and neural networks are; what an LLM is and how it is built (pretraining, fine-tuning, RLHF); and what tokens, context windows, embeddings, parameters, and multimodality mean. Explain *why* LLMs hallucinate and why they sound confident when wrong.

**D2 · Conversational skill (prompting).** Get reliably better output than a novice from the same model: give role, context, and constraints, provide examples, ask for structured output, iterate and critique, and know when to switch models or start fresh. Build and reuse a personal prompt library.

**D3 · The Claude platform.** Use Projects, Artifacts, Cowork, file uploads, Claude in Chrome, and connectors fluently; choose the right surface for a task; and set up a persistent Project that carries your thesis context across every chat.

**D4 · Literature discovery & synthesis.** Run an end-to-end literature workflow (discover with grounded tools, map a subfield, synthesize with Claude or NotebookLM, and manage references in Zotero) that produces a synthesis with **no fabricated or misattributed citations**, every claim traceable to a real paper you have read.

**D5 · Data analysis without coding.** Take a real lab dataset (CSV or Excel) and analyze it conversationally (cleaning, descriptive stats, appropriate basic tests, and publication-style plots), interpret the results critically, catch when the AI does something statistically wrong, and keep the work reproducible.

**D6 · Scientific writing & communication.** Use AI to outline, draft, tighten, and pressure-test scientific prose (manuscripts, aims, rebuttals, plain-language summaries) while staying within ICMJE and NIH policy: correct disclosure, no undisclosed authorship, no fabrication, and no AI in confidential peer review.

**D7 · Visual & media generation.** Produce clear scientific figures and schematics (BioRender), concept graphics for slides and posters (Gemini, Canva), and AI-generated decks (Gamma), efficiently and within a firm ethical line that never fabricates data-like imagery.

**D8 · Agents & automation (Claude Code + connectors).** Use the terminal at a basic level; run Claude Code to process folders of PDFs and data and automate research chores; understand agents conceptually; connect Claude to PubMed and Google Drive via MCP connectors; and know when an agent should *not* be trusted to act unsupervised.

**D9 · Responsible use & judgment.** Reason clearly about hallucination, bias, privacy, reproducibility, and research integrity; verify AI output as a reflex; explain to others what is and isn't appropriate; and make good build-vs-trust decisions. This is the connective tissue across all eight other domains.

## The self-assessment rubric

For each domain, you rate yourself against four levels. You start mostly at **Novice**. The plan is designed to carry every domain to **Proficient**, with several reaching **Local Expert** by the capstone. Re-score at each milestone.

| LEVEL | WHAT IT LOOKS LIKE (ANY DOMAIN) |
|---|---|
| **Novice** | Has heard the terms; uses AI like a search box; accepts output at face value; unaware of failure modes. |
| **Functional** | Completes the task with guidance or a template; gets useful results on simple cases; starting to spot obvious errors. |
| **Proficient** | Does the task independently on real work; structures prompts and workflows deliberately; verifies output; recovers when the model is wrong. |
| **Local Expert** | Fast and reliable on messy real cases; *teaches the workflow to others*; anticipates failure modes and designs around them; makes sound judgment calls about when not to use AI. |

**TARGET END-STATE PROFILE**

By the capstone, a realistic target is **Local Expert** in D1-D4, D6, and D9 (the daily research core) and **Proficient** in D5, D7, and D8, with clear paths to push those to expert as your own research demands. That profile meets the goal: the go-to person in your lab for applying AI to science and daily life.

SECTION 3

# Tools & budget

A single **$20/month Claude Pro** subscription is the engine for about 80% of this plan. Everything else is either free or added only for the weeks a phase needs it. The $200 budget is comfortable headroom rather than a spending target.

## 3.1 The Anthropic core

Research confirmed that the surfaces you need are bundled into consumer Claude plans. You do not need the API (pay-per-token, meant for building software) or any enterprise product.

| SURFACE | WHAT IT IS | PLAN | WHY IT MATTERS FOR YOU |
|---|---|---|---|
| Claude Pro | The core subscription: full model access, higher limits | $20/mo (≈$17/mo annual) | The anchor. Unlocks everything below. |
| Projects | Persistent workspace with custom instructions and uploaded knowledge every chat inherits | Included | One Project per paper, experiment, or grant keeps context loaded. |
| **Artifacts** | Live side-panel for documents, tables, small apps | Included | Draft docs, comparison tables, quick calculators. |
| **Cowork** | Desktop "AI coworker" that runs multi-step tasks over your files and folders | Included (Pro, since Jan 2026) | "Organize these data files" or "synthesize these PDFs" over a real folder. |
| Connectors (MCP) | One-click links to outside tools and data | Included, all plans | PubMed, Google Drive, BioRender, Benchling: the science set. |
| **Claude in Chrome** | Browser extension that can act on web pages | Included, paid plans (beta) | Navigate journal portals and databases; use cautiously. |
| Claude Code | Agentic tool that reads files, runs commands, edits, automates | Included (Pro/Max) | Phase 7: PDF and data wrangling, not software-building. |
| Claude for Excel | Spreadsheet integration for conversational analysis | Paid plans | Phase 4: analyze CSV or Excel without code. |

**THE “SCIENCE OFFERING,” ASSESSED HONESTLY**

Anthropic really does have a science push, and the practically useful part for you is already inside Pro:

- **Claude for Life Sciences** (launched Oct 2025): connectors into PubMed, Benchling, BioRender, and 10x Genomics, plus science “Agent Skills.” These work on a normal Pro plan. That is the practical, low-cost win, and it maps directly onto your wet-lab neuro workflow (literature, protocols, figures).
- **Claude Science** (the newer “AI workbench”): a desktop app oriented toward computational pipelines, packages, and compute, in macOS/Linux beta and leaning enterprise and dry-lab. You do not need it as a non-coding wet-lab scientist. Don’t budget for it, and revisit only if your research turns computational.

Some product and pricing specifics here were corroborated from official-page search snippets rather than loaded directly (vendor bot-blocking); see Section 10. None affect the recommendation: start on Pro.

## 3.2 When (and whether) to go beyond Pro

**Claude Max** ($100/mo for 5× usage; $200/mo for 20×) buys more usage and priority, not different core features. Start on **Pro** and step up to Max only if you repeatedly hit limits during heavy synthesis or data-analysis days, likely around Phases 3-4 or 7, if at all. It is a usage upgrade, not a prerequisite.

**Check first, might be free:** Vanderbilt may participate in **Claude for Education** or provide institutional access to paid tools (for example, Scite via the library, or Microsoft 365 Copilot). A five-minute check with your library or IT could make some of this free. As of mid-2026 there was no public self-serve .edu discount for consumer Claude; the education path is institution-licensed.

## 3.3 The full stack & a realistic monthly budget

| LAYER | TOOL | COST | WHEN |
|---|---|---|---|
| AI assistant, synthesis, writing, analysis, agents | **Claude Pro** | $20/mo | Always, from month 1 |
| Reference manager & citation gatekeeper | **Zotero** | Free | From Phase 3 |
| Grounded discovery & subfield mapping | **Semantic Scholar**, **ResearchRabbit** | Free | From Phase 3 |
| Grounded close-reading + audio overviews | **NotebookLM** (free tier) | Free | From Phase 3 |
| Evidence tables from real papers | **Elicit** (free tier, or Plus) | $0-12/mo | Phase 3, systematic-review weeks |
| “Did this finding hold up?” | **Scite** (check library first) | $0-12/mo | Optional, Phase 3 |
| Scientific figures & schematics | **BioRender** (academic/student) | ~$35/mo | Phase 6 & when making figures |
| Free diagramming | **draw.io**, **Excalidraw** | Free | Anytime |
| AI images for slides | **Google Gemini** (free tier) | Free | Phase 6 |

| | | | |
|---|---|---|---|
| AI slide decks | Gamma (free, or Plus $8-10) | $0-10/mo | Phase 6 |
| Design/posters + Claude connector | Canva (free / Education free) | Free* | Phase 6 |
| Optional usage upgrade | Claude Max | +$100/mo | Only if hitting limits |

*Canva for Education is free if eligible; otherwise a free personal tier covers basic needs. Prices are academic or student rates where available and were current as of July 2026; confirm at signup with a .edu email.

### TYPICAL MONTH

**$20** (Pro alone) for most months. A figure-heavy or systematic-review month might reach **$35 to $55** with BioRender and Elicit added. Even stacking Max on top stays under **$160**, below the cap with room to spare.

### THE ONE-LINE SPENDING RULE

Pay for Claude Pro always. Add a second paid tool *only* in the phase that uses it, and cancel it when that phase ends. Everything else in this plan is genuinely free.

SECTION 4

# The phased timeline

Nine phases run over about 26 weeks. Each lists its objectives, a short verified resource set, a required hands-on exercise on your own lab material, and the milestone that closes it. Every resource name is a live link, and costs are consolidated in Section 7.

### HOW TO READ EACH PHASE

**Learn** is the roughly 20-minute inputs. **Do** is the roughly 40-minute practice, always on real work. **Milestone** is the artifact you produce and self-score (Section 5) before advancing. Resource names in bold are clickable and are also detailed in Section 7.

PHASE 0 WEEK 1 (PART) · 2-3 HRS · DOMAINS D3, D9

# Setup & first contact

The goal of this short phase is zero friction: accounts live, tools installed, and one genuinely useful interaction, so momentum starts on day one. No terminal, no coding.

### Do

- Create or upgrade to **Claude Pro** ($20/mo). Install the **Claude desktop app** (Mac or Windows) and try **claude.ai** in the browser, with no command line involved.
- In account settings, review the privacy, memory, and training controls and set them to your comfort (see the data-hygiene box, Section 1).
- Make sure you have a Google account (for the free NotebookLM and Gemini used later).
- Start your AI Lab Notebook (a single Google Doc or Claude Project). Every session's notes go here.
- **First useful chat:** paste the abstract of a paper you're currently reading and ask Claude to explain the key finding, define three unfamiliar terms, and list two questions a skeptical reviewer would ask.

### Learn (light)

- Skim the **Anthropic Academy** catalog and bookmark it. These free official courses come back in Phases 2 and 7.

**✦ Checkpoint 0**

Accounts are live, the desktop app opens, and you've had one chat that actually helped you understand a paper. Your AI Lab Notebook exists with its first entry.

**PHASE 1** WEEKS 1-3 · 12-14 HRS · DOMAINS D1, D9

# Foundations: what AI & LLMs really are

Before using the tools deeply, you'll build an honest mental model of what's under the hood. As a neuroscientist you'll find the neural-network framing intuitive, and understanding why these systems hallucinate is what makes you a safe, critical user afterward.

### Objectives

Explain AI versus ML versus deep learning; what a neural network and an LLM are; pretraining, fine-tuning, and RLHF; tokens, context windows, embeddings, and parameters; and the mechanistic reason hallucinations happen.

### Learn (chunk the long videos across sessions)

- **3Blue1Brown**: "But what is a neural network?" (Ch.1, ~19 min) and the short "Large language models explained briefly" (~8 min). A clear visual intuition for the ideas. Save the transformer and attention chapters (Ch.5-6) for later in the phase.
- **Andrej Karpathy: "[1hr Talk] Intro to Large Language Models"** (~1 hr, split over 2-3 sessions). A strong conceptual map of training and the "LLM as OS or agent" idea.
- **Tim Lee & Sean Trott: "Large language models, explained with a minimum of math and jargon"** (long read). A cognitive scientist co-authored it, so the framing lands well for a neuroscientist: embeddings, transformers, training.
- **Structured anchor (pick one and run it through the phase): Elements of AI** (Helsinki, free) for a no-math "what is AI" spine, or **Generative AI for Everyone** (DeepLearning.AI, free audit) for the modern LLM-era version.

### Do

- After each video, write a 3-sentence explainer in your own words in your AI Lab Notebook. Teaching it back is the retention trick.
- Ask Claude to quiz you on the day's concept, then to grade your answers and correct any misconceptions.
- Deliberately catch a hallucination: ask Claude for five references on a niche neuro topic, then check whether they exist. They may not. This lesson underwrites all of Phase 3.

#### ✦ Milestone 1: "Explain an LLM"

Record a **5-minute talk** (phone camera or slides) explaining to a labmate what an LLM is, how it's trained, and why it hallucinates, all without notes. Self-score D1 on the rubric. Local-expert tell: you can field "so is it just autocomplete?" without hand-waving.

**PHASE 2** WEEKS 4-5 · 9-10 HRS · DOMAINS D2, D3

# Claude fluency & prompting

Now you turn the mental model into skill: getting consistently strong output, and setting up Claude to carry your research context. This is where the daily productivity gains begin.

### Objectives

Prompt deliberately (role, context, constraints, examples, structured output, iteration); use Projects, Artifacts, and Cowork; choose the right surface for a task; and build a reusable prompt library.

### Learn

- **Anthropic Prompt Engineering Interactive Tutorial** (official, free). Work the concepts; as a non-coder you can skip the code cells and keep the prompting lessons.
- **Anthropic Academy** prompting course (free, official) for a structured pass with a completion certificate.
- **Andrej Karpathy: "How I Use LLMs"**, a practical companion covering tool use, web search, and thinking models. Chunk it across sessions.

### Do (all on real lab work)

- Create a **Claude Project** for your thesis area: add custom instructions (your subfield, techniques, writing style) and upload 5-10 core papers and your lab's key protocols as project knowledge.
- Start a **prompt library** in your AI Lab Notebook: reusable prompts for "summarize this paper for my committee," "critique this paragraph," "turn these bullet points into methods prose," and "explain this technique to a rotation student."
- Use an **Artifact** to build a comparison table of three methods you're choosing between.
- Try **Cowork** on a folder: point it at a set of PDFs and ask for a one-paragraph summary of each.

#### ✦ Milestone 2: Thesis Project + prompt library

A working Claude Project loaded with your research context, plus **10 or more reusable prompts** you actually use. Demonstrate it by getting a genuinely useful, well-structured answer to a real research question in a single well-formed prompt. Self-score D2 and D3.

PHASE 3 WEEKS 6-9 · 18-20 HRS · DOMAINS D4, D9

# Literature review & synthesis

This is the highest-leverage research skill in the plan, and the one with the sharpest integrity risk. The core discipline: discovery happens in grounded tools that search real papers; Claude and NotebookLM synthesize only what you have actually retrieved; and Zotero is the gatekeeper for every citation.

### Objectives

Run an end-to-end literature workflow (discover, map, synthesize, and manage references) that produces a synthesis with no fabricated or misattributed citations, every claim traceable to a real paper you have read.

### Learn the tools (each under an hour; use the verified tutorials in Section 7)

| TOOL | ROLE IN THE WORKFLOW | COST |
|---|---|---|
| Zotero | Reference manager and citation gatekeeper; only real, retrieved papers live here | Free |
| Semantic Scholar | Free search backbone, TLDRs, and alerts | Free |
| ResearchRabbit | Visual citation-network mapping of a subfield; syncs to Zotero | Free |
| Elicit | Structured evidence tables from real papers (methods, n, outcomes) | Free / $12 |
| Consensus | Fast citation-backed check on yes/no empirical claims | Free / ~$10 |
| Scite | Did a finding hold up? Supporting versus contrasting citations | Library / $12 |
| NotebookLM | Grounded close-reading of papers you upload; cites every claim; audio overviews | Free |
| Claude (PubMed connector) | Synthesis, drafting, and reasoning across retrieved papers | In Pro |

### Do

- Pick a real question from your project (for example, “what’s known about [receptor/pathway] in [cell type] during [process]”).
- **Discover:** map the subfield in ResearchRabbit, pull an evidence table in Elicit, sanity-check claims in Consensus, and check whether key findings replicated in Scite. Import everything real into **Zotero**.
- **Synthesize:** load the retrieved PDFs into **NotebookLM** and a **Claude Project**, then ask each to summarize themes, agreements, and open questions, with every claim cited to a source you have.
- **Verify:** for every citation, confirm it exists and says what’s claimed (quote-search in the PDF or Google Scholar; check DOIs via Crossref; screen older references against **Retraction Watch**).

**THE NON-NEGOTIABLE RULE**

Never ask Claude (or any chatbot) to “give me references for X” from memory. It will produce real-looking citations that don’t exist, and fabricated-reference rates in the literature have risen sharply. Ask it to reason only over papers you supply, and let Zotero be the single source of truth for your bibliography.

### ✦ Milestone 3: Verified mini-synthesis

A **2-3 page synthesis** on a real question, drawing on about 10-15 papers, with a Zotero library and a short **verification log** proving every citation was checked, and zero fabricated or misattributed references. This is the plan’s integrity keystone. Self-score D4 and D9. Local-expert tell: you can teach the “discover, then synthesize, never invent” workflow to a labmate.

PHASE 4 WEEKS 10-12 · 14-15 HRS · DOMAINS D5, D9

# Data analysis without coding

You have no coding background, and you won't need one. Modern AI tools write and run the analysis code invisibly, so you work entirely in plain English. The skill to build is directing and checking the analysis, not writing it.

## Objectives

Take a real CSV or Excel dataset and analyze it conversationally (clean, describe, run appropriate basic tests, and make publication-style plots), interpret it critically, catch statistical mistakes, and keep it reproducible.

## Learn

- **Claude for Excel** (official support guide). This is your primary path, since Claude is your main tool.
- **OpenAI: "Data analysis with ChatGPT"** (official help doc), a clear reference for the upload-and-ask pattern. Optional, and only if you also keep a ChatGPT Plus seat.
- **Coupler.io: "Use Claude.ai for Data Analytics (Safely)"**. Read especially the privacy section on what not to paste.
- A preview of Phase 7: **VelvetShark: "Data analysis with Claude Code"** for datasets too big for chat upload.

## Do (on a real dataset from your rotation or lab)

- Upload a real spreadsheet; ask Claude to describe its structure, flag data-quality issues, and propose a cleaning plan, then approve each step.
- Run descriptive stats and generate a clear plot (bar with error bars, scatter, or time-course, as fits your data).
- Ask which statistical test is appropriate and why, then have it check the assumptions (normality, sample size, independence). Deliberately probe: ask a leading question and confirm it pushes back rather than agreeing.
- Have it produce a short, plain-language interpretation you could show your PI, and independently sanity-check one number by hand.

**STATISTICS GUARDRAIL**

AI will confidently run the *wrong* test if asked leading questions. Treat it as a fast analyst who needs supervision: make it justify each test, state its assumptions, and show its work. For anything headed toward publication, a real statistician or your PI still signs off. Never upload identifiable human data without IRB and data-use clearance.

### ✦ Milestone 4: Real dataset to stats and figure

Start from a raw lab spreadsheet and produce a **cleaned dataset, an appropriate statistical result you can defend, and a publication-style figure**, documenting the steps so it's reproducible. Self-score D5. Tell: you caught at least one thing the AI got wrong or oversimplified.

PHASE 5 WEEKS 13-15 · 14-15 HRS · DOMAINS D6, D9

# Scientific writing & research integrity

Writing is where AI saves the most time and carries the most policy risk. You'll learn to use it as a tireless editor and sparring partner, firmly inside the rules that govern manuscripts and grants.

### Objectives

Use AI to outline, draft from your own notes, tighten, and pressure-test scientific prose (manuscript sections, aims, rebuttals, plain-language summaries), with correct disclosure and no policy violations.

### Learn (read the policy first; it's load-bearing)

- **ICMJE Recommendations**: AI cannot be an author; all AI use in writing or editing must be disclosed; authors are fully responsible for every fact and citation; and reviewers must not upload manuscripts to AI tools.
- **NIH NOT-OD-23-149**: generative AI is prohibited in peer review, because uploading applications breaches confidentiality.
- **NIH "Apply Responsibly" (2025)**: applications substantially developed by AI aren't considered original, which is directly relevant to your future F30/F31 and grant writing.
- **ACS Nano best-practices** and the **MDPI good-practices** checklist, with concrete examples of what is and isn't OK.

**APPROPRIATE**

- Grammar, clarity, flow, and structure edits on *your own* draft
- Reordering and tightening a rebuttal letter you wrote
- Plain-language summaries and outlines
- "What would a reviewer attack here?" critique
- Disclosed use, per journal policy

**NOT APPROPRIATE**

- Generating substantive content and passing it off as yours
- Any fabricated data, figures, or citations
- Undisclosed use where disclosure is required
- Uploading confidential manuscripts under review
- Any AI use in NIH peer review

### Do

- Take a **real paragraph you wrote** and iterate it with Claude for clarity, keeping your voice and every claim your own.
- Draft a **Specific Aims** outline or a **methods section** from your own bullet points, then have Claude critique it as a study-section reviewer.
- Draft a **response-to-reviewers** letter for a paper (yours or a mentor's, with permission).
- Write a proper **AI-use disclosure statement** for a manuscript.

#### ✦ Milestone 5: Edited section + disclosure

A piece of real scientific writing you **materially improved with AI while staying within policy**, plus a correct disclosure statement and a one-paragraph note on where you drew the line. Self-score D6 and D9.

**PHASE 6** WEEKS 16-17 · 9-10 HRS · DOMAINS D7, D9

# Figures, slides & media

Claude has no image or video generation, so this phase is deliberately non-Anthropic, used either directly or through the Canva-Claude connector. The skill is producing clear scientific visuals quickly, inside a hard ethical line.

### Objectives

Make publication-grade schematics (BioRender), concept graphics for slides and posters (Gemini, Canva), and AI-generated decks (Gamma), efficiently and without ever fabricating data-like imagery.

**THE BRIGHT LINE (APPLIES TO EVERYTHING HERE)**

Never use any image or video tool to generate or "clean up" anything that could pass as experimental data, such as blots, micrographs, histology, flow plots, or IHC. That is fabrication, journals screen for it, and it ends careers. AI imagery is for **conceptual illustration and slide graphics only**, disclosed per journal policy.

### Learn & Do

- **BioRender** (student/academic plan) with its official Learning Hub: build a **real schematic** of your experimental design or a signaling pathway in your system. Its icons are vetted for scientific accuracy, which general image AI can't match. (For non-biological flowcharts, free **draw.io** or **Excalidraw** work fine.)
- **Google Gemini** (free): generate a concept or background graphic for a title slide. Strong photorealism and generous free access.
- **Gamma** (free, or Plus ~$8-10): turn your Milestone-3 synthesis or an outline into a **lab-meeting deck** in minutes, then refine.
- **Canva** (free or Education): optionally connect the verified **Canva MCP connector** to Claude and drive a poster or slide design from a chat.

**✦ Milestone 6: Figure + deck**

One **BioRender schematic** good enough for a real talk or paper, and one **AI-generated slide deck** for an actual lab meeting, produced in a fraction of your usual time. Self-score D7. Tell: you can state exactly why none of it crosses the data-fabrication line.

PHASE 7 WEEKS 18-22 · 22-25 HRS · DOMAINS D8, D9

# Claude Code, agents & connectors

The most ambitious phase, and the reason it runs five weeks: you've never used a terminal. The on-ramp is gentle and science-first. Think of Claude Code as a tool that reads a folder of files and does tedious work, not as a way to build software.

### Objectives

Use the terminal at a basic level; run Claude Code to process folders of PDFs and data and automate chores; understand agents conceptually; connect Claude to PubMed and Google Drive via MCP; and know when an agent should not act unsupervised.

### Week-by-week (gentlest to most capable)

| WK | FOCUS | KEY RESOURCES |
|---|---|---|
| 18 | Terminal primer: open it, navigate folders, run a command | **Anthropic Terminal guide**; **MDN command-line** (§1-3) |
| 19 | What an agent *is* (a chatbot answers; an agent acts, in a loop) | **IBM "What are AI Agents?"**; **explainX** beginner guide |
| 20 | Install & first Claude Code session (substitute science tasks for the code examples) | **Claude Code Quickstart**; **CC for Everyone: Non-Developers** |
| 21 | Claude Code for research: a folder of PDFs to themes and evidence tables | **CC for Everyone: Researchers**; **Academy: Claude Code in Action** |
| 22 | MCP & connectors: plug Claude into PubMed & Google Drive | **MCP intro** + **modelcontextprotocol.io** |

### Do

- Complete the terminal primer until "open a terminal, go to a folder, run one command" feels routine. (On Windows, Anthropic's native installer works, so WSL is not required.)
- Install Claude Code, and in your first session ask it to summarize a single PDF in a folder.
- **The core exercise:** point Claude Code at a folder of 10-20 papers or notes and have it produce a cross-file **evidence table or themed summary** with source filenames and verbatim quotes. (Two practical limits: scan-only PDFs need OCR first, and long PDFs are read in chunks.)
- Enable the **PubMed** and **Google Drive** connectors in Claude and run one real task through each.

**REFERENCE, NOT HOMEWORK**

Anthropic's **"Building effective agents"** is the canonical piece, but it's written for engineers. Skim only the opening definitions and the "workflows vs agents" distinction, and save the rest for if you ever go deeper.

### ✦ Milestone 7: PDF folder to evidence table

Using Claude Code, turn a folder of real papers into a **useful cross-document artifact** (a themed synthesis or evidence table with sources), and connect at least **one MCP connector** (PubMed or Drive) to a real task. Self-score D8. Tell: you can explain when you would not let the agent act without review.

PHASE 8 WEEKS 23-26 · 16-18 HRS · ALL DOMAINS

# Integration, capstone & the keep-current habit

The final phase chains every skill into one real research workflow, consolidates judgment, and sets up the lightweight habit that keeps you current after the plan ends.

### Learn (consolidate judgment)

- **Ethan Mollick, *Co-Intelligence*** (read across the phase): a clear narrative frame for working with AI as coworker, tutor, and coach, and for everyday and personal use (email, planning, learning).
- **MIT Sloan: "When AI Gets It Wrong"** and **AI Snake Oil / normaltech.ai**: the skeptical, integrity-minded counterweight to hype that every scientist should carry.
- Set up your **continuing-education habit** from Section 9: two podcasts and two newsletters, no more.

### Do: the capstone

Choose a real, small research question and run it **end to end, AI-accelerated**, documenting each step and every verification:

1. **Discover & synthesize** the relevant literature (Phase 3 workflow), citations verified.
2. **Analyze** a relevant dataset conversationally and make a figure (Phase 4).
3. **Build** a schematic and a short deck (Phase 6).
4. **Write** a 1-2 page brief with AI editing, within policy, with a disclosure note (Phase 5).
5. **Automate** one chore in the process with Claude Code or a connector (Phase 7).
6. Keep an **integrity and verification log** throughout (Phase 9 judgment).

### ✦ Milestone 8: Capstone research sprint

A single documented workflow that takes a real question from literature to a written, figure-supported brief, with every AI use appropriate and every citation and number verified. Re-score **all nine domains**. Meeting the target profile (Section 2) means you are locally expert. Final tell: a labmate could follow your write-up and reproduce the workflow.

**AFTER THE PLAN**

Fluency decays without use, and tools drift fast. Keep sharp by using these workflows in your actual research daily, keeping the Section 9 continuing-education habit, and revisiting your rubric each semester to push D5, D7, and D8 toward expert as your project demands.

SECTION 5

# Milestones & checkpoints

Eight milestones, one per phase, each a real artifact you produce and self-score, so “progress toward expert” becomes a tracker you fill in rather than a feeling. Print this page and use it at each mentor check-in.

| # | MILESTONE ARTIFACT | DOMAINS | “DONE” LOOKS LIKE | SELF-SCORE (N / F / P / LE) |
|---|---|---|---|---|
| **0** | Setup & first useful chat | D3 | Accounts live; one chat that helped | n/a |
| **1** | 5-min “explain an LLM” talk | D1, D9 | Explains training + hallucination, no notes | D1 ___ D9 ___ |
| **2** | Thesis Project + 10-prompt library | D2, D3 | One prompt → genuinely useful answer | D2 ___ D3 ___ |
| **3** | Verified mini-synthesis (10-15 papers) | D4, D9 | **Zero fabricated citations**; verification log | D4 ___ D9 ___ |
| **4** | Real dataset → stats + figure | D5, D9 | Defensible test; caught an AI error | D5 ___ D9 ___ |
| **5** | AI-edited section + disclosure | D6, D9 | Improved & within ICMJE/NIH policy | D6 ___ D9 ___ |
| **6** | BioRender figure + AI deck | D7 | Talk-ready; no data-like fabrication | D7 ___ |
| **7** | PDF folder → evidence table + connector | D8, D9 | Cross-doc artifact; 1 MCP connector live | D8 ___ D9 ___ |
| **8** | Capstone research sprint (end-to-end) | All | Reproducible; every use appropriate & verified | All ___ |

N = Novice, F = Functional, P = Proficient, LE = Local Expert (rubric, Section 2). Target end-state: LE in D1-D4, D6, D9; P in D5, D7, D8.

#### FAST CHECKPOINTS BETWEEN MILESTONES

- **Weekly (2 min):** did I do all my sessions on *real* lab material? Is my AI Lab Notebook current?
- **Per phase:** can I explain this phase’s skill to a labmate in 3 minutes?
- **The verification reflex:** did anything leave a chat and enter my science unchecked this week? (Target: never.)

#### WHAT TIPS A DOMAIN INTO “LOCAL EXPERT”

Two signals, both observable: (1) you **teach the workflow** to someone else and they succeed with it; and (2) you **predict the failure mode** before it happens and design around it. When both are true for a domain, mark it LE.

# Glossary of AI terms

Plain-English definitions, grouped so you can learn them in the order the plan introduces them. Skim it in Phase 1; return to it whenever a term appears.

## Foundations

### Artificial intelligence (AI)

The broad field of building software that performs tasks we associate with human intelligence: recognizing patterns, using language, making decisions. An umbrella term, not one technology.

### Machine learning (ML)

A branch of AI where systems *learn patterns from data* instead of following hand-written rules. Show it thousands of examples and it infers the rule itself.

### Deep learning

ML using neural networks with many layers (“deep”). It powers essentially all of today’s notable AI, including LLMs.

### Neural network

A model loosely inspired by brain neurons: layers of simple units connected by weighted links. During training the weights adjust so the network maps inputs to useful outputs. (The neuroscience analogy is loose: artificial “neurons” are far simpler than biological ones.)

### Parameters (weights)

The adjustable numbers inside a neural network: what it “learns.” Modern LLMs have billions. More parameters ≠ automatically better.

### Generative AI

AI that *produces* new content (text, images, audio, code) rather than only classifying or predicting a label.

### Large language model (LLM)

A very large neural network trained on huge amounts of text to predict likely next words. That single ability, at scale, yields writing, summarizing, answering, and reasoning. Claude, GPT, and Gemini are LLMs.

### Foundation model

A large model trained on broad data that can be adapted to many downstream tasks: the “base” others build on.

### Transformer

The neural-network architecture (introduced 2017) behind modern LLMs. Its key trick is *attention*. The “T” in GPT.

### Multimodal

A model that handles more than one kind of input/output: e.g., text *and* images. Claude can read an image of a figure and discuss it.

## How models are built & trained

#### Training

The process of adjusting a model's parameters by showing it data, so its outputs improve. Expensive and done by the model maker: not something you do.

#### Pretraining

The first, largest training stage: the model learns language and world patterns by predicting next tokens across an enormous text corpus. Produces a capable but unpolished "base model."

#### Fine-tuning

Additional training on a narrower dataset to specialize or align a model's behavior after pretraining.

#### RLHF (reinforcement learning from human feedback)

A fine-tuning method where humans rank model outputs and the model is trained to prefer the better ones: a big reason modern assistants feel helpful and follow instructions.

#### Alignment

The effort to make a model's behavior match human intentions and values: helpful, honest, harmless.

#### Training data

The text (and images, etc.) a model learned from. Its coverage and biases shape what the model knows and gets wrong.

#### Knowledge cutoff

The date after which a model has no built-in knowledge, because its training data ended there. Anything newer must come from search or documents you provide.

#### Benchmark

A standardized test used to compare models (e.g., on reasoning or coding). Useful but imperfect proxies for real-world usefulness.

## Model mechanics & internals

#### Token

The unit of text a model reads and writes: roughly a word-piece (~4 characters of English). Models "think" in tokens; pricing and limits are measured in them.

#### Context window

How much text (in tokens) a model can consider at once: the prompt plus its reply. Large windows let Claude hold many papers in mind simultaneously.

#### Embedding

A representation of text as a list of numbers (a vector) that captures meaning, so similar ideas sit near each other in "meaning space." The basis of semantic search.

#### Inference

Running a trained model to get an output (as opposed to training it). Each chat message is an inference.

#### Temperature

A setting controlling randomness/creativity in output. Lower = more focused and repeatable; higher = more varied.

#### Reasoning / “thinking” models

Models that spend extra computation working through a problem step-by-step before answering, improving hard multi-step tasks.

#### Chain of thought

Prompting or training a model to show intermediate reasoning steps, which often improves accuracy on complex problems.

## Using models: prompting & features

#### Prompt

What you send the model: your instructions, question, and any context. Prompt quality largely determines output quality.

#### Prompt engineering

The craft of writing effective prompts: giving role, context, constraints, examples, and a desired format.

#### System prompt

Background instructions that set a model’s persistent behavior for a session (e.g., a Project’s custom instructions).

#### Zero-shot / few-shot

Asking with no examples (zero-shot) vs. including a few worked examples (few-shot) to steer the output.

#### Context / grounding

Supplying source material (papers, data, docs) so the model answers *from that* rather than from memory: the key to trustworthy, citable output.

#### RAG (retrieval-augmented generation)

A pattern where the system first retrieves relevant documents, then has the model answer using them. Claude Projects and NotebookLM are RAG-style tools.

#### Project (Claude)

A persistent workspace with custom instructions and uploaded knowledge that every chat inside inherits: ideal for a thesis area.

#### Artifact (Claude)

A live side-panel workspace for a document, table, or small app the model builds and you can edit.

#### Cowork (Claude)

A desktop feature where Claude runs multi-step tasks over your files and folders in the background.

#### Canvas / workspace

General term (across tools) for an editable panel where AI-generated content is refined, as opposed to a linear chat.

## Agents, tools & connectors

#### AI agent

A system that doesn't just answer but *acts*: it plans, uses tools, observes results, and loops toward a goal. A chatbot answers; an agent does.

### Tool use / function calling

An LLM's ability to call external tools (search, a calculator, a database) to get real information or take actions beyond generating text.

### Claude Code

Anthropic's agentic tool that works in a terminal or editor: it reads files, runs commands, and edits. Useful for wrangling PDFs, data, and documents, not only for coding.

### MCP (Model Context Protocol)

An open standard, a "USB-C port for AI," that lets Claude connect to outside data and tools through a common plug instead of custom wiring.

### Connector

A one-click MCP integration linking Claude to a service: e.g., PubMed, Google Drive, BioRender, Canva.

### API (application programming interface)

A way for software to talk to a model programmatically, billed per token. Meant for building applications; a subscription is the right choice for an individual.

### Terminal / command line

A text interface for giving a computer typed commands. Basic familiarity is enough to run Claude Code; deep expertise isn't needed.

### Guardrails

Constraints that limit what an agent may do: important because an agent's mistake is a performed *action*, not just a wrong sentence.

## Research-specific

### Grounded search tool

A research tool (Elicit, Consensus, Scite, Semantic Scholar) that answers from a database of *real* indexed papers, so citations correspond to actual sources: far safer than a chatbot's memory.

### Smart citations (Scite)

Citation data showing whether later papers *support, mention, or contrast* a finding: useful for checking whether a result held up.

### Citation network

The web of which papers cite which; tools like ResearchRabbit map it visually to reveal a subfield.

### Reference manager

Software (e.g., Zotero) that stores papers and generates formatted citations. Because it holds only papers you actually added, it's a natural gatekeeper against fake references.

### Systematic review

A structured, comprehensive literature review with explicit search and screening criteria; several AI tools now assist the screening/extraction steps.

### NotebookLM

Google's free tool that answers strictly from sources you upload, citing each claim: strong for close-reading a fixed set of papers.

## Safety, ethics & limitations

### Hallucination

When a model produces confident, fluent output that is false: invented facts, fake citations, wrong numbers. It is a structural property of how LLMs work, not an occasional bug, so always verify.

### Confabulation

Another word for hallucination, borrowed from the neuroscience/psychology sense of filling gaps with plausible fabrication.

### Bias

Systematic skew in outputs reflecting biases in training data: relevant to fairness and to scientific interpretation.

### Sycophancy

A model's tendency to agree with the user or tell them what they want to hear: dangerous when you ask leading analytical or statistical questions.

### Data privacy / retention

Whether and how a service stores or trains on your inputs. Check settings before pasting unpublished or sensitive data; know your account's controls.

### Disclosure

Stating how AI was used in a manuscript or grant, as journals and funders increasingly require.

### Research integrity

Honesty and rigor in research. With AI this means: verify every fact and citation, never fabricate data or imagery, and follow authorship, disclosure, and peer-review rules.

### Reproducibility

Whether others can repeat your work and get the same result. Document AI-assisted analysis steps so they can be re-run and checked.

### Open weights / open source

Models whose parameters are publicly released (e.g., some Llama, Mistral models), which can run locally: relevant when data can't leave your machine.

### AGI (artificial general intelligence)

A hypothetical AI with broad, human-level competence across most tasks. Today's systems are narrow by comparison; the term is used loosely: treat bold claims skeptically.

# Resources used in this plan

Every resource the plan actually assigns, grouped by phase, each checked live in July 2026. The **Link** column is clickable. Costs are the individual/academic rate where one exists. See Section 10 for verification caveats.

## Foundations: what AI & LLMs are (Phase 1)

| RESOURCE | FORMAT | COST | LINK |
|---|---|---|---|
| 3Blue1Brown: Neural Networks series (incl. "But what is a neural network?", "LLMs explained briefly", transformers & attention) | Video | Free | 3blue1brown.com/topics/neural-networks |
| Karpathy: "[1hr Talk] Intro to Large Language Models" | Video ~1h | Free | youtube.com/watch?v=zjkBMFhNj_g |
| Karpathy: "Deep Dive into LLMs like ChatGPT" (reference/deep) | Video 3.5h | Free | youtube.com/watch?v=7xTGNNLPyMI |
| Lee & Trott: "Large language models, explained with a minimum of math and jargon" | Article | Free | understandingai.org/p/large-language-models-explained |
| Elements of AI: "Introduction to AI" | Course ~25h | Free | elementsofai.com |
| DeepLearning.AI: "Generative AI for Everyone" (Andrew Ng) | Course ~6h | Free audit | deeplearning.ai/courses/generative-ai-for-everyone |
| Ethan Mollick: *Co-Intelligence* (also used in Phase 8) | Book | ~$18 | penguinrandomhouse.com/books/741805 |

## Claude fluency & prompting (Phase 2)

| RESOURCE | FORMAT | COST | LINK |
|---|---|---|---|
| Anthropic: Prompt Engineering Interactive Tutorial | Course (GitHub) | Free | github.com/anthropics/courses |
| Anthropic Academy: courses catalog (prompting, Claude, Claude Code) | Courses | Free | anthropic.skilljar.com · anthropic.com/learn |
| Anthropic: Prompt engineering docs | Docs | Free | platform.claude.com/docs |
| Karpathy: "How I Use LLMs" | Video 2h | Free | youtube.com/watch?v=EWvNQjAaOHw |

## Literature review & synthesis (Phase 3)

| TOOL / RESOURCE | ROLE | COST | LINK |
|---|---|---|---|

| | | | |
|---|---|---|---|
| Zotero | Reference manager & citation gatekeeper | Free | zotero.org |
| Semantic Scholar | Free search backbone, TLDRs, alerts | Free | semanticscholar.org |
| ResearchRabbit | Citation-network mapping; Zotero sync | Free | researchrabbit.ai |
| Elicit | Evidence tables from real papers | Free / ~$12/mo | elicit.com |
| Consensus | Citation-backed yes/no claim checks | Free / ~$10/mo | consensus.app |
| Scite | Supporting vs contrasting citations | Library / ~$12/mo | scite.ai |
| NotebookLM | Grounded close-reading + audio overviews | Free | notebooklm.google |
| Retraction Watch (+ database) | Screen references for retractions | Free | retractionwatch.com |
| Tutorials: Elicit systematic-review guide; NotebookLM lit-review guide; ResearchRabbit guide | Tutorials | Free | elicit.com/blog/systematic-review · video |

## Data analysis without coding (Phase 4)

| RESOURCE | FORMAT | COST | LINK |
|---|---|---|---|
| Anthropic: “Use Claude for Excel” | Official guide | In Pro | support.claude.com/…/12650343 |
| OpenAI: “Data analysis with ChatGPT” (optional reference) | Official doc | Feature ~$20/mo | help.openai.com/…/8437071 |
| Coupler.io: “Use Claude.ai for Data Analytics (Safely)” | Article | Free | blog.coupler.io/how-to-use-claude-ai-for-data-analytics |
| VelvetShark: “Data analysis with Claude Code” (large files) | Article | Free | velvetshark.com/data-analysis-with-claude-code |

## Scientific writing & research integrity (Phase 5)

| RESOURCE | TYPE | COST | LINK |
|---|---|---|---|
| ICMJE Recommendations (AI, authorship, disclosure, peer review) | Policy | Free | icmje.org/recommendations |
| NIH: Generative AI prohibited in peer review (NOT-OD-23-149) | Policy | Free | grants.nih.gov/…/NOT-OD-23-149 |
| NIH: “Apply Responsibly” AI-in-applications policy (2025) | Policy | Free | grants.nih.gov → Apply Responsibly |

| | | | |
|---|---|---|---|
| ACS Nano: “Best Practices for Using AI When Writing Scientific Manuscripts” | Editorial | Free | pubs.acs.org/doi/10.1021/acsnano.3c01544 |
| MDPI: “Good Practices for Scientific Article Writing with… AI” | Article | Free | mdpi.com/2673-687X/3/2/9 |

## Figures, slides & media (Phase 6)

| TOOL / RESOURCE | USE | COST | LINK |
|---|---|---|---|
| BioRender (+ Learning Hub) | Scientific figures & schematics | ~$35/mo academic | biorender.com · /learn |
| draw.io / Excalidraw | Free flowcharts & sketches | Free | drawio.com · excalidraw.com |
| Google Gemini (image generation) | Concept/slide graphics | Free tier | gemini.google.com |
| Gamma (+ tutorial) | AI slide decks | Free / ~$8-10/mo | gamma.app |
| Canva (+ Canva-Claude MCP connector) | Posters, slides, design | Free / Education | canva.com · /help/mcp-agent-setup |

## Claude Code, agents & connectors (Phase 7)

| RESOURCE | LEVEL | COST | LINK |
|---|---|---|---|
| Anthropic: Terminal guide for new users | Absolute beginner | Free | code.claude.com/docs/en/terminal-guide |
| MDN: Command line crash course (§1-3) | Beginner | Free | developer.mozilla.org → Command line |
| IBM Technology: “What are AI Agents?” | Beginner video | Free | youtube.com/watch?v=F8NKVhkZZWI |
| explainX: “What Are AI Agents? A Plain-English Beginner’s Guide (2026)” | Beginner | Free | explainx.ai/blog/what-are-ai-agents… |
| Anthropic: Claude Code Quickstart | Beginner | Free | code.claude.com/docs/en/quickstart |
| Anthropic Academy: “Claude Code 101” → “Claude Code in Action” | Beginner → Int. | Free | anthropic.skilljar.com · Coursera |
| CC for Everyone: “Claude Code for Non-Developers” | Non-coder | Free | ccforeveryone.com/…/non-developers |
| CC for Everyone: “Claude Code for Researchers” | Non-coder | Free | ccforeveryone.com/…/researchers |
| Anthropic: “Introducing MCP” & the MCP intro docs | Beginner → Int. | Free | anthropic.com/news/mcp · modelcontextprotocol.io |
| Anthropic: “Building effective agents” (reference only; developer-level) | Advanced | Free | anthropic.com/engineering/building-effective-agents |

## Integration, judgment & responsible use (Phase 8)

| RESOURCE | TYPE | COST | LINK |
|---|---|---|---|
| Ethan Mollick: *Co-Intelligence* | Book | ~$18 | penguinrandomhouse.com/books/741805 |
| MIT Sloan: “When AI Gets It Wrong: Hallucinations & Bias” | Article | Free | mitsloanedtech.mit.edu/…/hallucinations-and-bias |
| Narayanan & Kapoor: AI Snake Oil / “AI as Normal Technology” | Book + essays | Free essays | normaltech.ai |

# If you have extra time

Optional depth: good resources that would over-stuff a one-hour-a-day plan but reward extra curiosity. None of it is required to reach the end state.

## Deeper on fundamentals & prompting

- **Karpathy: "Deep Dive into LLMs like ChatGPT"** (3.5h video), the complete training-stack reference if Phase 1 leaves you wanting more.
- **DeepLearning.AI: "ChatGPT Prompt Engineering for Developers"** (~1.5h, free). Hands-on prompting; it assumes a little Python, so it is optional for a non-coder.
- **Anthropic Courses on GitHub** (API fundamentals, tool use, evaluations) and the Anthropic Cookbook. Both are code-heavy, for if you ever want to peek under the hood.

## More literature-review firepower

- **Undermind**: agentic "deep" literature search with strong recall on narrow mechanistic questions; worth it during an intensive systematic-review push. Confirm current price at signup.
- **SciSpace**: all-in-one chat-with-PDF and formatting; check for a free university license.
- **Perplexity** **(student ~$10/mo)**: fast web-grounded orientation and background; always click through to sources, since it searches the open web, not just papers.
- **Zotero AI plugins**: PapersGPT (chat-with-PDF in your library; supports free local models), Aria, and Beaver.

## More image / media tools

- **Ideogram** (best text-in-image, free tier) for poster titles and labels; **Adobe Firefly** (commercially-safe training data) when licensing matters; and **Midjourney** for top aesthetics (paid, and more than most needs require).
- **AI video (free tiers):** Google Veo via Gemini; Pika and Kling for cheap experimentation; and Runway or HeyGen only if a specific project demands it. Video is a nice-to-have, not core.

## Worth checking with the institution

- **Claude for Education**: if Vanderbilt participates, it could make paid Claude access free. **Scite** and **Microsoft 365 Copilot** may also be library or IT-provided. Five minutes could save you real money.

# Continuing education: the keep-current habit

This field moves monthly, so staying current is a *light* habit rather than a second job. The list is deliberately short: **two podcasts and two newsletters** is the whole recommendation. Most of the rest is noise for a busy scientist.

## Podcasts: pick 2

| SHOW | WHY IT'S THE PICK | CADENCE | LINK |
|---|---|---|---|
| **Hard Fork** (NYT) | Top pick. Weekly, journalistic, entertaining, and jargon-free. A good "keep up without drowning" show for a non-coder. | Weekly | nytimes.com/column/hard-fork |
| **Google DeepMind: The Podcast** (Hannah Fry) | Strongest science fit. A mathematician interviewing researchers on AI in science, health, and biology, which matches your domain and level. | Fortnightly (by season) | deepmind.google/the-podcast |
| *Optional 3rd:* The AI Daily Brief | A roughly 20-minute daily current-events drip if you want more, and easy to skip when the hour is tight. | Daily | Apple Podcasts |

Cut for fit: Latent Space and Practical AI (too engineering-focused), Dwarkesh (strong but long and dense, a "graduate" option for later), No Priors (VC framing), and Lenny's (product management). Watch for Latent Space's spun-off "AI for Science" show, a promising domain fit, though not yet verified for cadence.

## Newsletters: pick 2-3 (all free)

| NEWSLETTER | WHY IT'S THE PICK | LINK |
|---|---|---|
| **One Useful Thing** (Ethan Mollick) | Top pick. Practical, non-technical, and experiment-driven. The most practical ongoing "how do I actually use this" source. | oneusefulthing.org |
| **The Batch** (DeepLearning.AI) | Weekly, well-curated by Andrew Ng's team, with balanced signal and little hype. | deeplearning.ai/the-batch |
| **AI as Normal Technology** (AI Snake Oil) | The skeptical, research-integrity voice: occasional deep reads rather than a weekly cadence. A useful counterweight for a scientist. | normaltech.ai |

Optional level-ups: Simon Willison's weblog (excellent, developer-leaning) and Import AI by Jack Clark (research-dense). Skip TLDR AI, which is high-volume and tooling-heavy.

## YouTube: optional, pick at most 2

- **Google DeepMind**: research explainers and the video podcast, on-domain.
- **AI Explained**: clear, analytical, non-hype breakdowns of why a model or trend matters.
- An alternative: **Two Minute Papers** for short, upbeat research summaries.

**THE HABIT, CONCRETELY**

One podcast on the commute or at the bench, one newsletter over coffee each week. That’s it. When a tool or model you use ships a major update, spend one session trying it on real work; hands-on beats reading about it.

SECTION 10

# Verification notes & caveats

Honesty about sourcing. This plan was researched by parallel specialist passes, each required to confirm that links resolve and content matches before inclusion. Here's what was solid and what to re-check.

## Method

Six research streams (Anthropic ecosystem; AI/LLM fundamentals; literature tools; Claude Code for non-coders; image/video/figure tools; and writing, productivity, and continuing-ed) were run and cross-checked against the live web on 8-9 July 2026. Two official Anthropic documentation pages were loaded in full; most other pages were confirmed via search results returning the page's own current text. Nothing dead, dormant, or unverifiable was knowingly included.

## Known limitations: re-check these before relying on a number

- **Vendor bot-blocking.** Several sites (anthropic.com, claude.com, YouTube, BioRender, Canva, some Substacks) return errors to automated fetchers. Their facts here are corroborated from official-page search snippets plus independent sources. Open them normally to confirm the exact figure.
- **Consumer-AI pricing shifts monthly.** All prices are July-2026 rates; confirm at signup, ideally with a .edu email for academic discounts (Elicit, Consensus, BioRender, Perplexity, and Scite all offer them, though exact post-discount numbers vary).
- **Claude plan to model mapping.** Which specific model each tier includes changes as Anthropic ships; the safe statement is that Pro and Max include the current Opus, Sonnet, and Haiku families.
- **Claude Science workbench.** Details (macOS/Linux beta, plan availability, any academic-lab discount) came from announcement snippets, not a loaded page. It isn't needed for this plan regardless.
- **Undermind pricing** and the **Latent Space "AI for Science" podcast** were flagged by research as not fully verifiable. Both are optional (Sections 8 and 9), so this doesn't affect the core path.
- **NIH and ICMJE policy is living guidance.** Always check the current version before a specific submission, since funder and journal AI rules are still evolving.

**TWO THINGS THAT DON'T CHANGE**

Whatever the prices do: first, never let any chatbot invent a citation, and verify every reference against a real source; and second, never generate imagery that could pass as experimental data. These are the load-bearing integrity rules of the entire plan.

# First-week checklist & starter prompts

## Week-one setup checklist

- ✓ Subscribe to **Claude Pro**; install the desktop app; open claude.ai.
- ✓ Review the privacy, memory, and training settings and set them to your comfort.
- ✓ Confirm you have a **Google account** (for NotebookLM and Gemini later).
- ✓ Create your **AI Lab Notebook** (one doc for prompts, wins, and errors).
- ✓ Bookmark **Anthropic Academy** and the Section 9 podcast and newsletter.
- ✓ Ask your library or IT whether Vanderbilt provides **Claude for Education** or Scite access.
- ✓ Do the first useful chat (below) on a paper you're actually reading.

## Starter prompts (copy, then adapt; wet-lab flavored)

**UNDERSTAND A PAPER**

*"I'm a neuroscience PhD student. Here is a paper's abstract [paste]. Explain its central finding in 4 sentences, define any specialized terms, list the main method used, and give two questions a critical reviewer would raise. Don't speculate beyond the text."*

**SET UP YOUR THESIS PROJECT (CUSTOM INSTRUCTIONS)**

*"You are assisting a Vanderbilt MD/PhD student doing basic-science neuroscience in a wet lab studying [system/technique]. Prefer precise, cautious, citable answers. When I ask about literature, only use sources I provide, and never invent citations. Flag uncertainty explicitly."*

**SYNTHESIZE PAPERS YOU UPLOADED (GROUNDED)**

*"Using only the PDFs in this Project, summarize what is known about [topic]. Group by theme, note where papers agree or disagree, and cite the specific source for every claim. List open questions. If something isn't in these papers, say so rather than guessing."*

**ANALYZE A DATASET (WITH PUSHBACK BUILT IN)**

*"Here is a CSV [upload]. Describe its structure and any data-quality issues. Propose a cleaning plan and wait for my approval. Then recommend an appropriate statistical test, state the assumptions it requires, and check whether my data meet them, and tell me if it's the wrong test rather than just running it."*

**EDIT YOUR WRITING (YOUR WORDS, YOUR CLAIMS)**

*“This is a paragraph I wrote for a manuscript [paste]. Improve clarity, flow, and concision without changing my meaning or adding any claims or citations. Keep my voice. Then, separately, tell me where a study-section reviewer would push back.”*

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Prepared as a personalized mentorship plan for Ezechukwu Nduka, July 2026. Anchored on Anthropic’s Claude, with non-Anthropic tools where they fit better. All resources were verified against the live web at the time of writing; confirm prices and policies at point of use. The two integrity rules, verify every citation and never fabricate data imagery, are non-negotiable.